

relationships with cloud providers and research partners. The second is disciplined experimentation through Quantum-as-a-Service, which provides access to different hardware modalities without premature capital expenditure. Candidate problems in molecular simulation, materials, scheduling, portfolio construction or grid optimisation should be tested through proofs-of-concept. Every result must be compared with strong, classical, high-performance computing and quantum-inspired baselines using total runtime, solution quality, energy and access cost, data-loading overhead, reproducibility and operational relevance. Only validated cases should move into the third horizon of pilots and hybrid production integration.

Investment decisions must therefore look beyond physical-qubit counts. More useful indicators include logical error rates, executable circuit depth, gate fidelity, throughput, compiler performance, availability, interoperability, and the cost of connecting quantum output to an existing decision process. Current vendor roadmaps place ambitious fault-tolerant systems near the end of the decade, but these remain engineering targets rather than guaranteed capacity. Diversification across hardware approaches, software layers and partners can reduce modality risk, vendor lock-in and dependence on a single roadmap while preserving intellectual-property options.

One investment can begin immediately: cryptographic modernisation. NIST's finalised post-quantum standards are available for implementation, and its migration programme emphasises cryptographic inventories, risk prioritisation, interoperability and benchmarking. Organisations should build cryptographic agility into procurement and architecture now, especially where data must remain confidential for

many years. The strongest quantum portfolio will ultimately combine measured experimentation with practical security work and invest in the interfaces through which quantum processors will cooperate with AI, cloud and supercomputing platforms.

Quantum as part of convergent technologies

Those interfaces place quantum within a convergent architecture rather than a separate computing silo. Quantum computing, quantum communication and quantum sensing must first be distinguished. Computing uses controlled qubits to execute specialised algorithms; communication distributes quantum states or keys across networks; sensing exploits quantum effects for precise measurements of time, fields or motion. Their maturity, infrastructure and use cases differ, although all three may eventually interact with AI, cloud platforms, high-performance computing, cybersecurity and advanced engineering.

A practical converged workflow will be heterogeneous. Data from laboratories, enterprise systems, digital twins or IoT sensors will be prepared by classical software. An orchestration layer will schedule work across CPUs, GPUs, supercomputers and, only where justified, a quantum processing unit. The QPU may evaluate a molecular state, sample a circuit or handle a formulated optimisation subproblem; a classical optimiser or AI model will interpret the measurements, update parameters, quantify uncertainty and return the result to the application. This loop matters because quantum machines require classical control, error decoding, compilation and post-processing, while data-encoding costs, latency and repeated measurements can

erase a theoretical advantage.

The infrastructure is already moving in this direction. EuroHPC's 2026 access programme allows European users to request several quantum technologies through the ecosystem used for supercomputers. Its LUMI-IQ initiative goes further by proposing an experimental platform combining HPC, AI and emerging quantum capabilities. Such developments make the quantum processor resemble a specialised accelerator within a shared computational fabric rather than a replacement for classical machines.

Convergence can also create compound capabilities. Peer-reviewed work on AlphaQubit showed that transformer-based machine learning can improve decoding of experimental quantum error-correction data, illustrating how AI may help stabilise future processors. In chemistry and materials discovery, quantum routines could estimate selected electronic properties, while HPC performs large-scale preprocessing and machine learning searches wider chemical spaces. In digital twins, a quantum optimiser might explore constrained alternatives while classical simulation verifies safety and feasibility. For 6G, robotics and distributed IoT, near-term quantum contributions may come more from sensing, timing and secure communications than from placing fragile QPUs at the edge.

Real value will depend on open source toolchains, common intermediate representations, portable APIs, workload schedulers, provenance and shared benchmarking. Post-quantum cryptography must protect the surrounding data and control plane, while quantum key distribution should be evaluated separately for specialised links. With these foundations, convergence becomes a disciplined method for assigning each technology the task it performs best.